



Lesson 6.6 — The Natural Logarithm

Big idea: $\ln$ is just log base e — the inverse of e^x . It's the log of choice in calculus, finance, and science because e is the base of continuous change.

Quick review

Definition: $\ln(x) = \log_e(x)$. Domain $(0, \infty)$, range $\mathbb{R}$.

Inverse pair: $\ln(e^x) = x$ and $e^{\ln x} = x$.

Same rules apply: product, quotient, power.

Identities: $\ln 1 = 0$, $\ln e = 1$.

Worked example

Worked example. Solve $\ln(x) + \ln(x - 3) = \ln(10)$.

$\ln[x(x - 3)] = \ln(10) \Rightarrow x(x - 3) = 10 \Rightarrow x^2 - 3x - 10 = 0 \Rightarrow (x - 5)(x + 2) = 0 \Rightarrow x = 5$ (reject -2 because of domain).

Warm-ups

1. Evaluate $\ln(1)$.
2. Evaluate $\ln(e)$.
3. Evaluate $\ln(e^3)$.

Practice

1. Solve $\ln(x) = 2$ for x (exact answer).
2. Solve $e^x = 10$ for x (exact and decimal).
3. Expand $\ln(x^2y/z^3)$.

Challenge

1. Solve $3e^{2x} = 24$.
2. Solve $\ln(x + 2) - \ln(x) = \ln 5$.



Answer key — Lesson 6.6

Check your work. If a problem tripped you up, rewatch the step in the video where that concept appears.

Warm-ups

1. 0.
2. 1.
3. 3.

Practice

1. $x = e^2$.
2. $x = \ln 10 \approx 2.3026$.
3. $2 \ln x + \ln y - 3 \ln z$.

Challenge

1. $e^{2x} = 8 \Rightarrow 2x = \ln 8 \Rightarrow x = (\ln 8)/2 = \ln(2^3)/2 = (3 \ln 2)/2 \approx 1.0397$.
2. $\ln[(x + 2)/x] = \ln 5 \Rightarrow (x + 2)/x = 5 \Rightarrow x + 2 = 5x \Rightarrow x = 1/2$.